

1) Quadratics

a) Solving by factoring

Example 1

Solve a) $6x^2 + 11x - 35 = 0$
 b) $20x^2 + 80 = 82x$.

Always look to see if we can first divide each term by a common factor.

a) $6x^2 + 11x - 35 = 0$
 $(3x - 5)(2x + 7) = 0$
 $x = \frac{5}{3}$ or $x = -\frac{7}{2}$ ← Solve $(3x - 5) = 0$ and $(2x + 7) = 0$.

b) $20x^2 + 80 = 82x$
 $20x^2 - 82x + 80 = 0$ ← Rearrange as $ax^2 + bx + c = 0$.
 $10x^2 - 41x + 40 = 0$ ← Divide through by any integer common factor.
 $(5x - 8)(2x - 5) = 0$
 $x = \frac{8}{5}$ or $x = \frac{5}{2}$

Example 2

A rectangle has length $(x + 3)$ metres and width $(2x + 1)$ metres.
 The area of the rectangle is 12 m^2 .
 Find the length of the rectangle.



Area = $(2x + 1)(x + 3) = 2x^2 + 7x + 3 = 12$
 $2x^2 + 7x - 9 = 0$
 $(2x + 9)(x - 1) = 0$
 $x = -\frac{9}{2}$ or $x = 1$

But the solution cannot be $x = -\frac{9}{2}$ as this will lead to a negative length and width, so $x = 1$.
 Length = $x + 3 = 1 + 3 = 4$ metres.

b) Solving inequalities

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|------------------------------|-------------------------------|---------------------------------|
| 2. a) $x^2 + 7x + 12 \leq 0$ | b) $x^2 - x - 30 < 0$ | c) $x^2 + 2x - 48 > 0$ |
| d) $x^2 - 5x + 6 \geq 0$ | e) $2x^2 - 11x + 12 > 0$ | f) $3x^2 + 14x - 5 \leq 0$ |
| g) $15 + 2x - x^2 \leq 0$ | h) $16 - 6x - x^2 < 0$ | i) $21 - x - 2x^2 > 0$ |
| j) $10x^2 + 43x + 28 \geq 0$ | k) $x^2 - 9 \geq 0$ | l) $50 - 2x^2 < 0$ |
| 3. a) $x^2 - 2x > 35$ | b) $x^2 + 6 \leq 5x$ | c) $x^2 \leq x + 20$ |
| d) $x(x + 3) \geq 10$ | e) $x^2 < 9x + 10$ | f) $x^2 - 4(x + 6) > 21$ |
| g) $24 > 11x - x^2$ | h) $7 + 2(4x^2 - 15x) \leq 0$ | i) $\frac{x^2 + 12}{2} \geq 4x$ |
| j) $(x + 5)^2 > 1$ | | |

c) Solving by quadratic equation

Example 13

Solve the inequality $2 - 6x - x^2 \leq 0$. Leave your answer in surd form.

Consider $2 - 6x - x^2 = 0$

$$x = \frac{-(-6) \pm \sqrt{(-6)^2 - 4(-1)(2)}}{2(-1)}$$

$$= \frac{6 + \sqrt{44}}{-2} \text{ or } \frac{6 - \sqrt{44}}{-2}$$

$$= \frac{6 + 2\sqrt{11}}{-2} \text{ or } \frac{6 - 2\sqrt{11}}{-2}$$

$$= -3 - \sqrt{11} \text{ or } -3 + \sqrt{11}$$

Now $2 - 6x - x^2 \leq 0$

$$x \leq -3 - \sqrt{11} \text{ or } x \geq -3 + \sqrt{11}$$

$$a = -1$$

$$b = -6$$

$$c = 2$$

$$\sqrt{44} = \sqrt{4 \times 11} = 2\sqrt{11}$$

**Example 14**

Solve the equation $2x^2 = 5x - 1$. Write your answer correct to 2 decimal places.

When asked to solve to 2 decimal places, it is likely that the quadratic formula will be appropriate.

$$2x^2 - 5x + 1 = 0$$

$$a = 2, b = -5, c = 1$$

$$x = \frac{-(-5) \pm \sqrt{(-5)^2 - 4(2)(1)}}{2(2)}$$

$$= \frac{5 + \sqrt{17}}{4} \text{ or } \frac{5 - \sqrt{17}}{4}$$

$$= 2.28 \text{ or } 0.22 \text{ (2 d.p.)}$$

We must first rearrange to $ax^2 + bx + c = 0$.

Q In examinations, unless otherwise indicated, answers to all questions on all topics should be rounded to 3 significant figures.

However, do **not** round your answers to 3 significant figures until the final answer.

A significant number of candidates lose marks in examinations by rounding earlier in the question which then leads to an inaccurate answer.

You should therefore keep all numbers unrounded until the final answer.

Exercise 1.4

1. Solve each of these quadratic equations. Write your answers correct to 2 decimal places.

a) $2x^2 - 3x - 4 = 0$ b) $5x^2 - 11x + 4 = 0$ c) $3x^2 + 12x + 5 = 0$ d) $x^2 + 5x - 2 = 0$

e) $4x^2 + 2x - 5 = 0$ f) $x^2 + x - 4 = 0$ g) $3x^2 = 6x - 2$ h) $9x = 6x^2 + 2$

i) $3 - 8x = 2x^2$ j) $x^2 + 3x = 1$

2. Use the formula to solve each of these quadratic equations. Leave your answers in surd form.

a) $2x^2 - x - 5 = 0$ b) $3x^2 - 6x + 1 = 0$ c) $5x^2 - 3x - 7 = 0$ d) $6x^2 - x - 4 = 0$

e) $x^2 + 8x - 3 = 0$ f) $x^2 + 7x + 3 = 0$ g) $x^2 + 10x - 5 = 0$ h) $2x^2 + 5x - 4 = 0$

i) $4x^2 = 5 - 10x$ j) $4x - 1 - x^2 = 0$

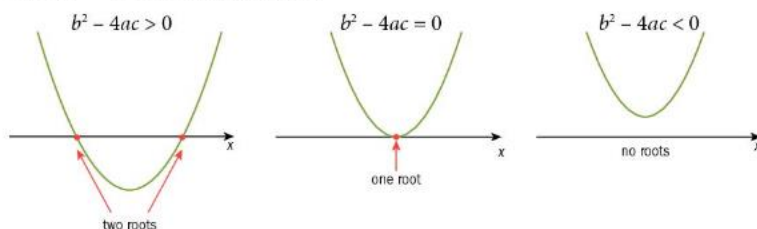
3. Use the formula to solve the following inequalities. Leave your answers in surd form.

d) Discriminant of quadratic equation

Looking at the discriminant of the quadratic equation

The discriminant is the value of $b^2 - 4ac$ in $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

- If $b^2 - 4ac > 0$ there are two distinct **real roots**.
 If $b^2 - 4ac = 0$ there are equal roots (one repeated root).
 If $b^2 - 4ac < 0$ there are no real roots.



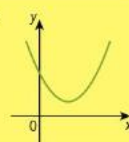
Example 18

Work out whether each of these quadratic equations has two distinct roots, equal roots or no real roots.

- a) $x^2 - 3x + 5 = 0$
 b) $3x^2 + x - 6 = 0$
 c) $25x^2 + 20x + 4 = 0$

Note: We know from 1.3 that if $y = ax^2 + bx + c$ and $a > 0$, then the curve is $\cup$ shaped.

If $b^2 - 4ac < 0$, then there are no real roots and the curve does not cross the x -axis, so y is positive for all values of x .



- | | |
|--|--------------------|
| a) $b^2 - 4ac = (-3)^2 - 4(1)(5) = 9 - 20 = -11 < 0$ | no real roots |
| b) $b^2 - 4ac = (1)^2 - 4(3)(-6) = 1 + 72 = 73 > 0$ | two distinct roots |
| c) $b^2 - 4ac = (20)^2 - 4(25)(4) = 400 - 400 = 0$ | equal roots |

- Show that $1 + 100x^2 - 20x \geq 0$ for all values of x .
- If $2x^2 - ax + 8 = 0$ has no real roots, find the range of possible values of a .
- If $6 - 2x - kx^2 = 0$ has a repeated root, find the value of k .
- The equation $x^2 + px + q = 0$, where p and q are constants, has roots -3 and 2 . Find the value of p and q .
- Use the discriminant to find the nature of the roots of the equation $3x + 4 = \frac{5}{x}$.
- The quadratic equation $kx^2 + 5x + 2 = 0$ has two distinct real roots. Find the range of possible values of k .
- Find the value of p for which the quadratic equation $px^2 - 4px + 2 - p = 0$ has equal roots.
- Prove that the quadratic equation $(q - 5)x^2 + 5x - q = 0$ has real roots for any value of q .
- The quadratic equation $x + k + \frac{9}{x} = 0$ has equal roots. Find the two possible values of k .
- The equation $px^2 + qx + r = 0$, where p , q and r are constants, has roots $-\frac{1}{2}$ and $\frac{3}{4}$. Find the smallest possible integer values of p , q and r .

e) Graphs of quadratic functions

Example 23

Express $y = x^2 - 6x + 10$ in the form $y = (x + p)^2 + q$.

Sketch the curve, stating the coordinates of the vertex.

$$y = (x - 3)^2 - 9 + 10 = (x - 3)^2 + 1$$

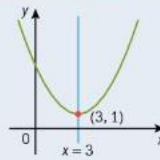
Coordinates of the vertex are (3, 1).

The **minimum** value of y is 1.

$a > 0$ therefore the graph is $\cup$ shaped.

The line of symmetry: $x = \frac{-b}{2a} = \frac{-(-6)}{2(1)}$

$$x = 3$$



$$\begin{aligned} b^2 - 4ac &= (-6)^2 - 4(1)(10) \\ &= -4 < 0 \\ &\text{therefore no real roots.} \end{aligned}$$

Example 24

Express $y = 3 - 8x - 2x^2$ in the form $y = a(x + b)^2 + c$.

Sketch the curve, stating the coordinates of the vertex.

$$\begin{aligned} y &= -2[x^2 + 4x] + 3 \\ &= -2[(x + 2)^2 - 4] + 3 \\ &= -2(x + 2)^2 + 8 + 3 \\ &= -2(x + 2)^2 + 11 \end{aligned}$$

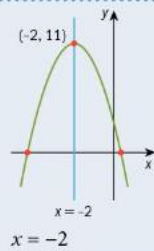
Coordinates of the vertex are (-2, 11).

The **maximum** value of y is 11.

$a < 0$ therefore the graph is $\cap$ shaped.

The line of symmetry: $x = \frac{-b}{2a} = \frac{-(-8)}{2(-2)}$

$$x = -2$$



$$\begin{aligned} b^2 - 4ac &= (-8)^2 - 4(-2)(3) \\ &= 88 > 0 \\ &\text{therefore two distinct roots.} \end{aligned}$$

Example 25

A sheep pen is in the shape of a rectangle.

One of the sides of the pen is a wall.

A farmer puts fencing on the other three sides of the pen.

The two sides that touch the wall are each x metres.

He uses 40 m of fencing.

Find the maximum area of the pen.



The third side of the pen = $40 - 2x$

$$\begin{aligned} \text{Area of the pen} &= x(40 - 2x) \\ &= 40x - 2x^2 \\ &= -2[x^2 - 20x] \\ &= -2[(x - 10)^2 - 100] \\ &= -2(x - 10)^2 + 200 \end{aligned}$$

The **maximum** area is 200 m² (when $x = 10$).

Summary:

To solve quadratic equations

- Factorise and solve.
- Or use the **method of completing the square**.
- Or use the quadratic formula $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$.

To solve quadratic inequalities

- Solve the corresponding equation $ax^2 + bx + c = 0$ to find the values of x .
- Sketch the graph of $y = ax^2 + bx + c$.
- Write down the solutions using inequality signs.

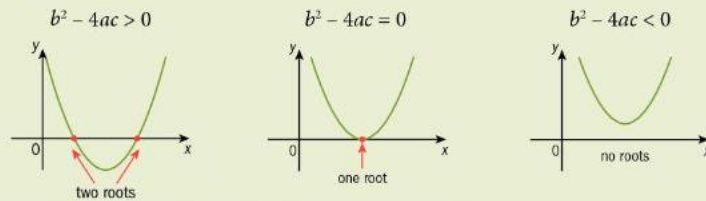
The discriminant of the quadratic equation $ax^2 + bx + c = 0$

- Discriminant $= b^2 - 4ac$

If $b^2 - 4ac > 0$ there are two distinct real roots.

If $b^2 - 4ac = 0$ there are equal roots (one repeated root).

If $b^2 - 4ac < 0$ there are no real roots.



$ax^2 + bx + c = 0$, where a , b and c are constants, $a \neq 0$.

Graphs of quadratic equations $y = ax^2 + bx + c$ where a , b and c are constants, $a \neq 0$

- If $a > 0$, the curve is $\cup$ shaped and has a **minimum** value.
- If $a < 0$, the curve is $\cap$ shaped and has a **maximum** value.
- The **equation of the line of symmetry** is $x = \frac{-b}{2a}$.

To solve simultaneous equations where one is linear and one is quadratic

- Rearrange the linear equation to write x in terms of y or y in terms of x .
- Then substitute this into the quadratic equation.

Recognise and solve equations in x which are quadratic in some function of x

- e.g. To solve $x^6 - 2x^3 + 1 = 0$, let $y = x^3$ and solve $y^2 - 2y + 1 = 0$.